# CHAPTER 6

# INTELLECTUAL PROPERTY AND TECHNOLOGY STARTUPS: WHAT ENTREPRENEURS TELL US

Stuart J. H. Graham and Ted S. Sichelman

## ABSTRACT

*This chapter provides evidence on how young technology startups are employing intellectual property (IP) protection when innovating and competing in the United States. Although researchers and teachers of university technology transfer often think only in terms of patents and the Bayh-Dole Act, this chapter suggests that adopting a more nuanced view of IP rights is appropriate. After reviewing the primary non-patent types of IP protection available in the U.S. (copyright, trademark, and trade secret), we explain that while patents are often considered the strongest protection, for some entrepreneurs – particularly those operating in the U.S. software and Internet sectors – patents may be the least important means of capturing value from innovation. We present evidence from the 2008 Berkeley Patent Survey to demonstrate that IP is used by U.S. startups in very different ways, and to different effects, across technology sectors and other company-specific characteristics. Contrary to the common assumption in academic discourse, we show*

**Technological Innovation: Generating Economic Results**
**Advances in the Study of Entrepreneurship, Innovation and Economic Growth, Volume 26, 163–199**
**Copyright © 2016 by Emerald Group Publishing Limited**
**All rights of reproduction in any form reserved**
**ISSN: 1048-4736/doi:10.1108/S1048-473620160000026006**

*that different forms of IP protection often serve as complements, rather than substitutes.*

**Keywords:** Copyrights; trademarks; trade secrets; patents; entrepreneurship

## INTRODUCTION

When considering intellectual property (IP) protection in the context of technology commercialization and university actors, those studying the phenomenon often think first of the Bayh-Dole Act,[1] and of patents. University patenting of federally funded inventions was made substantially less burdensome by the Bayh-Dole Act of 1980, and the proliferation of university technology-transfer offices – and the regime of disclosure and licensing under which many universities operate today – is largely built upon patents and the process of patenting. However, completely understanding the commercialization of university knowledge assets is incomplete without addressing other non-patent types of IP protection. These protections play an important role in the successful commercialization of many technologies, protecting the entrepreneur when taking the technology from idea, to proof-of-concept, to the market. Without comprehending these non-patent protections, we cannot have a complete appreciation of the incentives that drive risk-taking and success in the commercialization game.

The purpose of this chapter is to elaborate upon these non-patent forms of IP, and to discuss how technology entrepreneurs use the various IP protections in their menu of options. We argue in this chapter that, while researchers and teachers of university technology transfer often think exclusively in terms of patents and the Bayh-Dole Act, we ought to adopt a more nuanced view of the role played by IP rights for technology entrepreneurs. Accordingly, while patents are normally the "default" position when we think about technology startups protecting and profiting from innovation, we should develop a more comprehensive view of the IP environment, and the opportunities facing technology entrepreneurs.

In addition to introducing the primary non-patent types of IP protection (copyright, trademark, and trade secret), we present new evidence from U.S. technology startups generated from a survey about their executives' experiences using the different "appropriability" options available to them, and how and under what circumstances entrepreneurs find the methods

useful in competition. These findings, derived from the *2008 Berkeley Patent Survey* (BPS) described in Graham, Merges, Samuelson, and Sichelman (2009), are striking: They show that various strategies differ greatly in their reported effectiveness, these differences driven mainly by the underlying technology and other characteristics of the executives (people) and company (firm). These findings support prior research (Cohen, Nelson, & Walsh, 2000; Levin, Klevorick, Nelson, & Winter, 1987) discussed elsewhere in this volume (Ceccagnoli & Rothaermel, 2016, Chapter 1) but also diverge in important respects. As such, the BPS results reported in this chapter are particularly relevant for teaching technology entrepreneurship, since the respondents to this survey are top executives (CEOs and CTOs) at technology startups (biotechnology, medical devices, IT hardware, and software/Internet) founded less than 10 years before the survey was administered in 2008.

The responses of startup executives to the BPS described in this chapter also support the notion that IP and other means of appropriating value from technology are commonly complementary to one another. A common preconception is that different IP protections often act as substitutes to one another. We point out that a more comprehensive view is appropriate, and recommend broadening the lens from the single "invention" to a more complete view of innovation and commercialization as a dynamic and complex process (Sichelman, 2010).

Following Graham and Somaya (2006), we can consider strategic, contemporaneous uses of different types of IP. Complements in a purely economic sense are goods that tend to be consumed together, and BPS responses to questions about the importance of these different methods of appropriating value are positively correlated (tend to move together). The BPS results thus support the notion that entrepreneurs are examining their value strategy in a dynamic fashion, using patents, copyrights, trade secrets, and trademarks together throughout the commercialization process in ways that appear to complement one another. Accordingly, such complementary strategies may offer a greater protection and more opportunities to capture value from technology products or services for the entrepreneur.

The balance of this chapter is organized as follows: the following section introduces the three major types of non-patent legal protection available to entrepreneurs: copyright, trademark, and trade secret, and some discussion of why we should care about them, particularly in the context of technology commercialization. In the next section, we discuss the limitations of the IP-as-substitutes view, suggesting that our perspective ought to be expanded – particularly in this era of cumulative and complex

technologies – to consider IP and non-IP protections as complementary to one another. Then, we summarize evidence from the BPS (Graham et al., 2009), dealing particularly with the results of how different IP protections and other non-legal appropriability methods (lead time, difficulty of reverse engineering, and employing complementary assets like marketing or manufacturing) are viewed by startup executives, analyzing why startups commonly forgo patents on their major innovations. Finally, we conclude by discussing how these issues add to our understanding and may be used to influence our teaching of technology commercialization.

## USING IP TO APPROPRIATE VALUE FROM INNOVATIONS

Since Schumpeter (1934), we have understood that the innovation process includes much more than mere "invention." Innovation encompasses all of the elements necessary to bring a new technology or product to market, such as attracting capital, building the needed complementary resources, and achieving success in the marketplace. When considering the profits that may be captured from an innovation, we are often interested in an ongoing stream of "supernormal" profits, those returns that can be sustained and protected ("isolated") from being quickly competed away by imitation, entry, and the operation of otherwise normal market forces.

Recapping Ceccagnoli and Rothaermel in this volume (Chapter 1), scholars describing the environment faced by innovators trying to secure such profits typically speak of the "appropriability regime" (Rumelt, 1984; Teece, 1986). Teece (1986) suggests that the ability of innovators to capture profits is influenced by two forces – the technological characteristics of the good and the existing legal environment (largely in terms of IP protection). In terms of the former factor, goods that are easily replicable provide few barriers to competitors, while characteristics that are hidden (e.g., manufacturing processes) offer hurdles difficult for competitors to overcome. When considering the legal environment, stronger rights in and more active enforcement of IP (patents, trademarks, copyrights, and trade secrets) can provide advantages to innovators attempting to keep competitors at bay and secure for themselves supernormal returns.

Because patenting is addressed at length elsewhere in this volume (Rector, Sandefur, Ceccagnoli, Clendenin, & Hallenborg, 2016, Chapter 4; Holbrook, 2016, Chapter 5), we address here the relationship between

technology commercialization and the non-patent forms of IP. In the United States, the principal types of IP protection available in addition to patent include copyright, trademark, and trade secret protections. While this chapter will discuss only these three types at length, other protections are available under state common law (such as misappropriation)[2] and statutes (such as under the Semiconductor Chip Protection Act)[3] that can offer other valuable safeguards for innovators' commercialization activities.

## *Copyright Protection*

Copyright protection in the United States is largely governed by a federal statute.[4] Copyright protects original expressions, not ideas or processes, so long as the expression is fixed in some concrete, tangible medium, such as a document, a disk, or film.[5] Allowable expressions under the act may be literary, musical, dramatic, graphic, audio-visual, or architectural, among others.[6] The protection granted is against others who engage in unauthorized copying, distribution, or display of the work; copyright has a duration equal to the life of the author plus 70 years, or, in the case of works created for hire, the shorter of 95 years from publication or 120 years from creation.[7]

Unlike patent protection, it is not necessary that the expression be novel, useful, or non-obvious – rather, it need only satisfy a modicum of originality and not be copied from another. Because the protection begins from the moment that the expression is fixed in a tangible medium (e.g., as pen is put to paper), there is no formal registration requirement. Registration is, however, available from the U.S. Copyright Office after deposit of a copy of the work with the Library of Congress. Such registration is relatively inexpensive ($35 for single authors),[8] and affords the copyright holder certain legal advantages, including a presumption of validity (if registered within 5 years of original publication),[9] the option of filing an infringement suit (which may not be brought absent registration),[10] and enhanced infringement damages.[11]

Copyright protection became more relevant in the context of technology commercialization after 1980, when software "writings" became allowable subject matter. During the 1970s, copyright protection was promoted by policymakers as the preferred protection for software inventions (Menell, 1987). In its 1979 report, the National Commission on New Technological Uses of Copyrighted Works (CONTU), charged with making recommendations to Congress regarding software protection, endorsed copyright as the most appropriate form of protection for

computer software (CONTU, 1979). Because copyright protection attaches to an author with relative ease and has a long life – now upwards of 125 years for works created for hire – the Commission determined that copyright was the preferred type of IP protection for software (Samuelson, 1984). Congress adopted the Commission's position when it wrote "computer program" into the Copyright Act in 1980, and computer databases and programs are considered protected "literary works" under the Act.[12]

One cannot understand the application of copyright to software "writings" without understanding that patent protection was extended to software "ideas" at roughly the same time (Graham & Mowery, 2003). The federal courts' initial interpretation of copyright protection as applied to software suggested strong sweeping protection for inventors.[13] Although the development of court precedent over time has somewhat weakened this sweep (while patent protection for software was becoming more generally accepted in the 1990s),[14] copyright remains an important protection for software innovators. In fact, previous survey evidence suggests that software engineers prefer to rely upon copyright protection as compared to patent protection, partly on philosophical grounds (Samuelson & Glushko, 1990). The U.S. Supreme Court's 2014 decision in *Alice Corp. v. CLS Bank Int'l*, which effectively restricted the scope of what software could be patented, has further dampened the benefits to patenting software.[15] Given the ease, speed, and low cost of securing a copyright as compared to a patent, the latter costing from $20,000 to $60,000 on average for young firms to prosecute (Graham et al., 2009), copyright offers distinct advantages to software entrepreneurs with limited time and financial resources.

In the university context, the treatment of copyrightable materials varies widely. The University of Washington, for example, specifically provides in its policy documents that certain copyrightable materials are property of the university, subject to restrictions for "scholarly works" (University of Washington, 2007). Anecdotal discussion with officers at the University of California's technology-transfer office suggests that certain universities, like Washington, have historically been relatively aggressive at trying to capture value on copyrightable materials from their faculty. Others, like California, have been almost completely inactive.

### *Trademark Protection*

Trademark protection is not an IP protection per se. Unlike patent and copyright protections, which emanate from the U.S. Constitution's IP

Clause and are intended to incentivize creators, trademarks emanate from the Constitution's Commerce Clause and are commonly viewed as protecting consumers from fraud. That said, the reality is that trademark has become a de facto form of IP, and is useful to innovators in protecting marketing capabilities, brands, and goodwill. These assets may be among the most valuable held by private firms (WIPO, 2013).

Trademarks, like copyrights, are protected in the United States primarily under federal law. Nevertheless, important common-law elements of trademark exist in the individual states, and the current federal regime is largely a validation of these states' rights. As such, common-law trademark rights can offer valuable protection against sales of competing products that lead to consumers' confusion or mistake as to the origin, quality, or advertising of the protected goods.[16]

Federal law provides for the registration of marks used in commerce, and federal protection is predicated upon successful registration.[17] Registered marks may be trademarks, service marks, or other types of marks providing information about the origin of a commercial good, service, or related activity.[18] The registration process is overseen by the United States Patent and Trademark Office (USPTO), which requires that the mark be owned, be used – or intended to be used – in commerce, and that the applicant believes both that the mark is not in use, and is not confusingly similar to another mark in use.[19] Successful completion of the registration process leads to the mark being recorded on the Principal Register, serves as notice to all other later users of confusingly similar marks in the United States, and can earn for the mark the status of being incontestable.[20] Thus, registration can convert state-based common-law rights into a truly strong and national IP protection.

Given that the underlying policy motivation for trademark protection is to protect the consumer, to be valid the mark must be distinguishable from other marks. Marks are considered distinguishable either if they (1) have the characteristic of being arbitrary or are, at most, suggestive of the product, or (2) are descriptive of the product, but have acquired "secondary meaning" after prolonged commercial association. Marks that are merely "generic" are never capable of protection.[21] The mark "Intel" for microprocessors is considered suggestive of the intelligence or intellect needed to produce or consume the product, while the mark "Old Crow" for distilled whiskey may be considered arbitrary since it does not suggest or describe a significant characteristic of the good.

The protection given to the owner of a distinctive mark is to prevent others from employing confusingly similar marks, and to allow the owner

to use the federal (or state) courts to enforce that limitation on competitive uses. An owner often enjoys a broad scope of protection around the mark's unique value, since courts can stop others from employing different marks that tend to associate different goods of lower quality with the trademark owners products.[22] Trademark law has thus developed to give the holder of the mark valuable geographic protections that prevent others from using marks that either tend to confuse consumers or dilute the value of the mark – protections that offer entrepreneurs an important shelter from competition when engaging in distribution, sales, and marketing.[23]

### *Trade Secret Protection*

Although trade secret protection in the United States is derived mainly from state law, the passage of the Defend Trade Secrets Act (DTSA) will likely expand federal protection. Trade secrets have a set of more-or-less general characteristics, regardless of the jurisdiction in question. The subject matter of trade secrets is "valuable information," and thus differs from both patent and copyright. The information must generally be commercial information that affords some advantage or has independent economic value derived from its secrecy.[24] Moreover, the information cannot be readily ascertained by proper means. In order to keep the protection, the holder of the trade secret must also take reasonable steps to ensure that the valuable information remains a secret – thus, many companies demand non-disclosure agreements of their employees, and take care to institute policies on the treatment of company secrets.

The legal remedy available to the holders of a trade secret is to prevent others from misappropriating the valuable information. "Misappropriation" derives from the tort law, and as such, the concept generally involves some morally suspect behavior, such as stealing the information, or enticing persons to disclose the information wrongfully.[25] Unlike patent protection, courts will not intervene to prevent another's use of the secret information if the information was not "misappropriated." Therefore, if the valuable information was independently discovered or legitimately reverse engineered, trade secret law offers no remedy.

Like trademarks, trade secrets in theory have no limit to their duration. So long as the information remains valuable, the holder takes reasonable steps to keep the information secret, and nobody reverse engineers or independently discovers the secret, there is no limit to the protection's life. Furthermore, the protection may extend not only to "positive" knowledge

(e.g., customer lists) but also to "negative" knowledge (e.g., a process that was tested and found *not* to work) – both may provide advantage or value, and so are capable of trade secret protection.

## PATENT AND NON-PATENT PROTECTIONS: SUBSTITUTES OR COMPLEMENTS?

As stated earlier, scholars have tended to consider patenting in isolation. When the relationship between different types of IP has been considered, these protections have often been viewed as substitutes for one another. So, the question is often posed to the entrepreneur: Is patent *or* copyright more appropriate for protecting software? Or, at what point is it appropriate to *stop* trade secret protection and *start* patent protection?

The view that different types of IP act as substitutes for one another is common in much of the management and economic literature (Graham & Somaya, 2006). Because IP rights are creatures of the law, and because economists began analyzing IP protection quite early (Plant, 1934a, 1934b), it is not surprising that these disciplines had a strong influence on later notions of the relationships among types of IP. In the law, the substitutes view has been supported by court precedent. In *Kewanee Oil v. Bicron Corp.* (1974),[26] the U.S. Supreme Court declared that trade secrets protected "lesser or different inventions" than did patents. Economists Friedman, Landes, and Posner (1991), in an analysis of the economics of trade secret use, provide a strong rebuttal to the Court's notion that patents are preferred for "better" inventions, arguing that the likelihood of imitation, life of the invention, and relative costs of patenting and secrecy determine the choice. Nevertheless, these and other scholars continued to advance the notion that it is natural to think in terms of substituting patent for trade secret protection.

This IP-as-substitutes view has been strongly influenced by the importance of patent and trade secret protection in the early decades of the 20th century, driven by their widespread use in important turn-of-the-century industries like chemicals and machinery. Because the patent laws require disclosure, it is natural to think that the trade secret would be extinguished when a patent application is filed. Driven largely by this view, theoretical studies of appropriability have analyzed the economics of this tradeoff (Arora, 1995; Friedman et al., 1991; Horstman, MacDonald, & Slivinsky, 1985).

But this view of patents and trade secrets as substitutes for one another considers only the individual "invention." In today's world, the innovation process has many layers and often involves complex technologies, with potentially thousands of individual "inventions" embodied in a single product. Any handheld device, with telephone, Internet browsing, camera, and a variety of "apps," is but one example of such complexity. The concepts of "cumulative innovation" and "complex technologies" have entered our lexicon, and they suggest that the commercialization of technology is dynamic, in terms of both time and space. If we move away from the single "invention" to a complex invention and the entire innovation process, it becomes apparent that different types of IP may act in a complementary fashion (Graham & Sichelman, 2008, 2016; Graham & Somaya, 2006). Accordingly, these different mechanisms may bring benefits to the entrepreneur simply through their coincident use.

This way of thinking is not new among firms. For instance, Pilkington, PLC employed both patents and trade secrets in ways that complemented each other during the 1960s–1970s. Pilkington, the innovator of the "float glass" manufacturing process that revolutionized pane glass production – and in so doing lowered costs substantially – used both patents and secrecy in its appropriation strategy.[27] After gaining patent protection on the "float glass" process, Pilkington engaged in worldwide licensing, giving licensees exclusive territories but including a knowledge grant-back provision in the contracts. Pilkington closely guarded its trade secrets on process innovations (tacit knowledge not included in the patent documents), and after the patents expired in the early 1980s the company continued to collect royalties under licenses based around its perpetual trade secrets.[28]

Results reported from an analysis of the Carnegie-Mellon Survey in Cohen et al. (2000) lend support to the notion that different appropriability mechanisms may be complementary in use. The authors report correlations of industry-level mean effectiveness scores of the various surveyed mechanisms (e.g., lead time, secrecy, patenting), demonstrating that in the case of process technologies, the use of patenting and secrecy are positively correlated. Moreover, their factor analyses show that, in some circumstances, secrecy loads with patenting, leading the authors to suggest that there may be a premium to keeping to-be-patented innovations secret until the patent actually issues (Cohen et al., 2000). Another explanation may be that trade secrets are being used as a complement to patents.

Although patents and their implications for firm strategy, innovation, and commercialization have been treated elsewhere in this volume (Ceccagnoli & Rothaermel, Chapter 1; Holbrook, 2016, Chapter 5), we

stress that patents have long been a primary focus of researchers in management, economics, and law when analyzing the commercialization of university technology. This "patent focus" may be a result of the controversy within the university community over Bayh-Dole (Mowery & Sampat, 2001, 2004)[29] and the emergence of many university technology-transfer offices in response to the new regime. Outside the university setting, research into technology commercialization and R&D outcomes has also often been focused on patenting. Drivers of that focus include the fast-paced growth in patenting over the last two decades (Hall, 2005), primarily in the information and biotechnology sectors, and the relative ease for researchers of obtaining and using patent data (Griliches, 1990; Hall, Jaffe, & Trajtenberg, 2001). This treatment in scholarship – made less costly by increasing computing power – has contributed to patents often being considered in isolation, and as a sort of "gold standard" of IP protection.[30]

While we certainly do not support the position of scholars advocating for an abandonment of the patent system altogether (Boldrin & Levine, 2013), it is enough to make the point that entrepreneurs commonly use a diverse set of different appropriation methods. In fact, evidence from innovations originating in different European countries during the mid-1800s (when some countries, like Germany, recognized patent protection while others, like Switzerland, did not) shows that different types of protections can support innovation, but that without a patent system in the menu of options available to entrepreneurs, nations tend to produce different types of innovations (Moser, 2005). In the U.S. innovation system, patents have been shown to be of varying perceived importance (Cohen et al., 2000; Levin et al., 1987) across different technologies and industries (for a full discussion, see Ceccagnoli & Rothaermel, Chapter 1 of this volume).

## SURVEYING ENTREPRENEURS: THE 2008 BERKELEY PATENT STUDY

Although a considerable body of previous work has explored the relationship between IP rights and innovation, far less scholarship has focused on the more particular relationship between IP rights and entrepreneurship. The basic economics justifying IP rights recognizes that developing innovative products and practices is an expensive, time-consuming, labor-intensive, and risky endeavor (Arrow, 1962). Once these innovations exist,

however, they can be cheap and easy to copy. Having IP rights protects innovators from copying by "free riders" and allows them to recoup the investment incurred during the creation, development, and commercialization process – either directly by manufacturing and distributing products and services embodying the innovation, or indirectly through licensing to other firms incorporating the innovation in their products and services (Arora, Fosfuri, & Gambardella, 2004). This basic economic principle applies to entrepreneurial companies as well as to other firms in the marketplace. Moreover, because early-stage firms tend to lack the kinds of "complementary" assets (such as well-defined marketing channels, manufacturing capabilities, and access to credit), they are arguably even more sensitive to IP rights than larger firms (Teece, 1986). Nevertheless, research and policy analysis has not adequately addressed how particular IP laws differentially affect entrepreneurial firms relative to more established ones (Sichelman & Graham, 2010a).

In response to this noticeable gap in knowledge, we conducted a wide-scale survey in 2008 of high-technology startup firms in the United States to determine how these companies use – and are affected by – patents and other forms of IP (Graham et al., 2009). Part of our aim in conducting the survey was to identify those aspects of the patent system that encourage or hinder entrepreneurship, particularly in high-growth technology industries (the Internet, computer software and hardware, medical devices, and biotechnology).

Our survey results show that a substantial share of startup firms, concentrated especially in the software industry, avoid the patent system altogether. At the same time, we find that startups hold more patents and patent applications than prior commentators have reported. These differences appear to be primarily attributable to the completeness of our survey method, since our approach captures firm-owned patents and applications – including those acquired from founders and from other sources – which previous studies had missed.

A dominant theme in our findings is that the use and usefulness of patents to technology entrepreneurs is driven by the industry or sector in which they compete. Our results also shed light upon startups' motivations for patenting – and choosing not to patent – their major innovations. When electing to patent, startup executives tend to be most influenced by a desire to prevent copying and, to a lesser extent, by reputational and financial motives, including both accessing capital and successful exit (such as being acquired or going public). But the explanation for startups choosing not to patent is also context specific: for instance, biotechnology company

executives are much more likely to cite concerns about technology disclosure than those in other industries.

### *Surveying Technology Startups*

In the BPS we surveyed "entrepreneurial companies" – defined by us as U.S. companies founded less than 10 years prior. Noting the almost complete lack of hard data on what entrepreneurs actually *do* when faced with decisions about IP, we targeted our survey questionnaire to CEOs, presidents, and chief scientists or CTOs of technology startups. Wanting to report on companies most likely to generate innovations and growth, we selected two groups to survey: those representing the "population" of firms (drawn from Dun & Bradstreet – D&B)[31] and a smaller slice funded by venture capital (VC) investment (drawn from Thomson's *VentureXpert* database – VX).[32] Wanting also to explain innovation performance among startup companies, we limited our inquiry to "high-technology" sectors, including companies competing in the biomedical (biotechnology or medical devices) and information technology (software/Internet or hardware, the latter defined as semiconductor, communications, and computers) segments.[33]

In profile, our "median" respondent (at the 50th percentile) is a self-described "startup" company founded in April 2002 that has not yet successfully "exited" (through acquisition or IPO). The company has nine employees, half of whom are scientists or engineers, with 2007 revenues of $300,000, and founders who had prior experience running another company. Geographically, the company's offices are located somewhere west of the Mississippi river. In terms of funding, the "median" responding D&B company received funding from "friends and family," at least one "angel" investor, and from a commercial bank, but not from VC, investment banks, or other companies.

Over 1,300 unique companies responded to our survey. We find that, by and large, our respondents are not statistically different from the non-respondents (those who were contacted, but who chose to not answer) on key company characteristics. Within industries, we find generally no reportable difference in the age, sales volume, and employee counts between respondents and non-respondents. We also recognized that more active patentees may have selected into our "patenting" survey, so we matched our respondent and non-respondent companies to the USPTO patent records for patent grants and applications assigned to company names. While such patent matching is difficult (Thoma et al., 2010), we find

no significant differences[34] between the two groups in terms of the number of patents held by the companies, nor in the number of patent applications published.

### *Technology Entrepreneurs' Approach to Profiting from Innovation*

A major finding of the BPS is that while patenting plays an important, critical role for many high-technology startups, its importance is not ubiquitous and is often industry specific. While previous surveys generally failed to distinguish among important legal mechanisms, such as copyrights and trademarks (Cohen et al., 2000; Levin et al., 1987), our questionnaire asked the respondent to indicate how important (or unimportant) the following seven methods were to the company in securing competitive advantage from its technology innovations: (1) first-mover advantage over competitors; (2) secrecy; (3) patents; (4) copyrights; (5) trademarks; (6) difficulty of reverse engineering; and (7) other production, implementation, or marketing capabilities. Hereafter, we call these seven methods "appropriability strategies."

Our results show that these early-stage firms use multiple appropriability strategies. Among all respondents, first-mover advantage is ranked the most important. In fact, it is the only appropriability strategy ranked between "moderately important" and "very important" on average by companies in each technology category. Three methods are grouped together in the next ranked position – secrecy, complementary assets, and patenting – with respondents rating each on average between "slightly important" and "moderately important." Following these in rank are the remaining three appropriability strategies – difficulty of reverse engineering, trademarks, and copyright – which fall between "slightly important" and "moderately important."

Lumping all respondents together allows us to give a basic descriptive picture of our results. First-mover (lead time) advantage tends to dominate the other strategies, while other methods – including patents, secrecy, copyright, and trademark – are generally rated as having some importance. Apart from first-mover advantage, the differences between the average scores are relatively small (even though some of these differences are statistically significant). While lumping the responses together provides a quick look, it also hides substantial differences in the way respondents answered, especially in different industries and when pursuing different innovation approaches (Graham & Sichelman, 2011).

In fact, when we examine companies' responses by industry, the importance of these appropriability strategies tends to shift radically (Fig. 1). In biotechnology, medical devices, and even IT hardware firms in our sample, patenting is ranked highly, but not to the exclusion of other methods. Among biotechnology companies, patenting is ranked the most important appropriability strategy. For medical device startups and venture-backed IT hardware companies, respondents rank patenting second, behind "first-mover advantage" (lead time).

The value of patenting among startups in biotechnology and medical devices (and venture-backed IT hardware) stands in stark contrast to the (un)importance ascribed to patents by software and Internet firms.

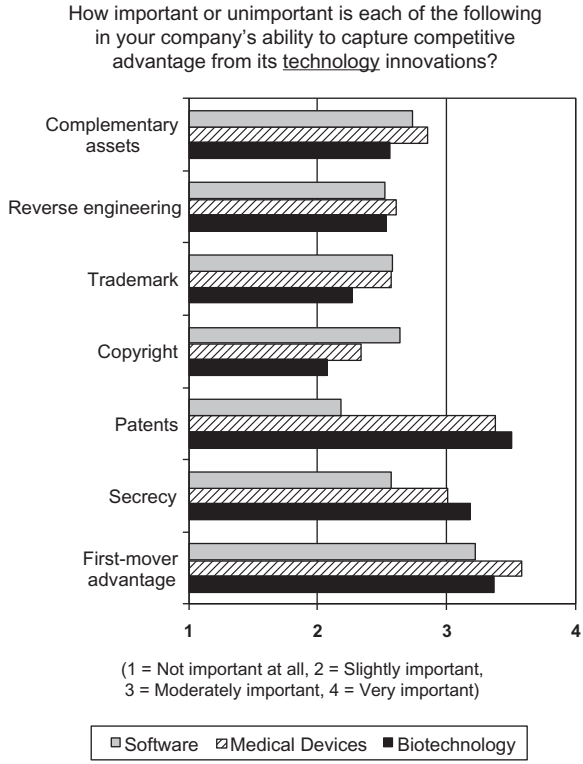



*Fig. 1.* Capturing Competitive Advantage from Technology, by Industry. Average of All Responses within Industry Reported. *Source*: Reproduced from Graham et al. (2009).

The limited function served by patenting in technology competition for early-stage software firms is underscored in Fig. 1. Among software entrepreneurs, patenting is rated the least important among all the appropriability strategies. When we focus only on software companies in the D&B sample, patenting is still the least important method, ranked on average barely above "slightly important." Likewise, patenting is the lowest-rated method by venture-backed software companies, although for these companies it is not statistically different from other low-ranked methods.

Our BPS results also allow us to separate companies based on their innovation focus, and we find that "process innovators" are substantially less likely than "product innovators" to report that patenting is important to them in capturing competitive advantage. To relate appropriability strategies with the innovation focus of early-stage companies, we asked our respondents to disclose the importance (or unimportance) of several types of innovation to their overall business strategy, specifically (a) product innovation; (b) process or internal tools innovation; (c) business-model innovation; and (d) design innovation (including product shape and packaging). For two categories, *product* and *process* innovation, we were able to identify companies considering one or the other as their firm's primary innovation focus.[35]

Our analysis demonstrates that, in general, patenting is ranked as relatively unimportant among process innovators compared to product innovators. In fact, patenting is rated as second only to first-mover advantage by product innovators, but is rated last out of all the methods by process innovators. First-mover advantage, secrecy, copyright, trademark, and the difficulty of reverse engineering are also rated as more important by product than by process innovators, but not to the same extent as the difference in the relative importance of patenting reported between the two types of innovators. Specifically, product innovators rate patents almost twice as important as do process innovators. Of all the methods, only complementary assets is ranked (in absolute terms) as a more important appropriability strategy by process innovators when compared with product innovators.

Noting that biotechnology companies rated patenting as more important overall than did software and Internet firms, we also examined product innovators in each of these two technologies. A strong technology effect on the ranking of patents' importance is once again evident: biotechnology product innovators are much more likely to rate patents as important when compared with software product innovators. In fact, among these product-focused biotechnology companies, patenting remains the most important

appropriability strategy, while among product-focused software companies, patenting remains rated the least important means of successfully competing. Therefore, we believe that our main findings are driven more by technology (sector) differences than innovation type (i.e., *process* vs. *product*).

### *Technology Entrepreneurs Use Multiple Methods to Protect Innovation*

As was suggested in the aggregated statistics, Fig. 1 demonstrates that startups across the high-technology landscape use different methods to compete with their technologies. While patenting is playing a substantial role in all but the software and Internet sector, startups in all sectors use other appropriability strategies – possibly in complementary ways (Sichelman & Graham, 2010a). It is noteworthy, however, that software and Internet startups tend to rate all of the methods as less important than companies in other sectors. Software and Internet companies rate on average only one method, first-mover advantage, as highly as "moderately important." Conversely, companies in other industry sectors rate at least three methods on average as at least "moderately important" (Fig. 1).

While biotechnology firms are active patent-holders and are more likely to say that patents are important to their competitive position, they also tend to rank several, if not all, of the various means of "capturing competitive advantage" from technological innovation more highly relative to software companies. Moreover, in order to effectively compete, startups across all industries tend to report that multiple methods of appropriability are useful. However, how different firms employ, and the relative importance they ascribed to, these methods do not follow a common pattern.

For instance, in terms of startups' use of IP, it is noteworthy that both copyright and trademarks play varied appropriability roles. These forms of protection are particularly salient for executives at software firms, although even in this sector there is divergence. Among the "population" of software companies (D&B), executives rank copyright as second, not statistically different from complementary assets (but both ranked behind first-mover advantage, which is ranked significantly as the most important method). For these same software-firm respondents, trademarks are ranked just behind copyright and complementary assets, and are considered just as important as using secrecy in capturing competitive advantage. Among the venture-backed software firms, however, trademarks, copyright, and patenting rank behind all others, statistically undifferentiated among each other as the least important methods of capturing competitive advantage.

The responses of D&B software companies differ markedly not only from venture-backed software firms, but also from companies operating in the other technology sectors. Among non-software startups, copyright is rated the least important of the several methods, while trademarks tend to be among the lower-ranked items.

In sum, while we find that various appropriability strategies are important to technology startups, our chief finding is that, outside of the software and Internet sector, patenting plays an important role in helping early-stage technology companies compete, but is by no means the only form of protection entrepreneurs are using to secure profits. Moreover, firms appear to be using the different forms of protection together. These findings raise a set of important questions: What are the specific mechanisms by which this "competitive advantage" is achieved? Is competitive advantage attained through added financing that protections facilitate, thus enabling the entrepreneur to develop a better technology and get it to market faster and more effectively than its competitors? Or is this competitive advantage won when patents and other protections act as "signals" to suppliers and would-be customers that a company has a valuable and important technology? Or, do protections permit the company to secure their innovations and keep competitors at bay while developing a competitive position that can be sustained, protected, and solidified? We could answer these questions only by inquiring into the specific factors motivating companies in seeking protections on their innovations.

### *Technology Entrepreneurs' Motives for Patenting*

We find that when technology entrepreneurs seek patent protection, they often do so for varying and often complementary reasons. Our review of the literature and extensive interviews with financiers, entrepreneurs, and executives suggested several factors motivating startups to seek patent protection. These included: (a) preventing others from copying products or services; (b) improving the chances of securing investment; (c) obtaining licensing revenues; (d) improving the chances/quality of liquidity (e.g., acquisition/IPO); (e) preventing patent infringement actions against the company; (f) improving the company's negotiating position with other companies (e.g., cross-licensing); and (g) enhancing the company's reputation/product image (Graham & Sichelman, 2008). In the BPS, we asked whether the company had filed at least one U.S. patent, and among

those answering in the affirmative we inquired into the importance of these motivations for seeking patents.

Across respondents, the most important reason for patenting is to prevent others from copying the startup's products and services (Fig. 2). This motivation remained paramount for cohorts segmented by a variety of characteristics, including dataset sampled (D&B or VX), sector, age, patenting intensity, and total revenues (Sichelman & Graham, 2010a). This result is notable because some previous commentators had opined that the high costs of patenting and enforcing patents generally precluded startups from using patents to prevent copying and competition (Mann, 2005; Meurer, 2008). While our evidence suggests that startups are not precluded from accessing the system, the respondents did indicate that costs were nevertheless a primary barrier to using the system (see Figs. 4–7 below and accompanying discussion).

Our respondents reported that financing and improving valuation upon exit – such as an acquisition or IPO – played a moderately to very important role in their decision to file for patents (Fig. 2). These results help

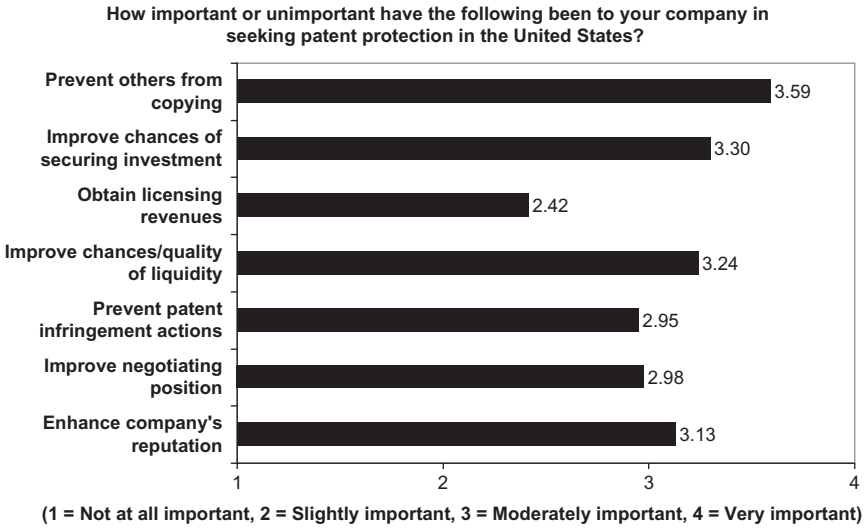



*Fig. 2.* Motivations for Patenting – All Startups Filing for U.S. Patents. The Above Question was Asked of Those Reporting that Their Company had Filed for at Least One U.S. Patent (Averages Reported). *Source*: Reproduced from Graham et al. (2009).

explain why startups seek patents, and contrast with some large-firm surveys in which respondents ranked patenting for securing capital as relatively unimportant. However, this finding is consistent with some information in the Carnegie-Mellon Survey (Cohen et al., 2000) suggesting that the relatively smaller firms tended to rank the importance of patenting to enhance firm reputation higher than the relatively larger companies in their sample. Our findings on the financing value of patenting are also consistent with studies showing that patenting plays a positive role in valuation during fundraising and upon exit for venture-backed firms (Cockburn & MacGarvie, 2009; Hsu & Ziedonis, 2007).

Like preventing copying, our financing result is robust across a variety of firm characteristics, including industry, age, patenting intensity, revenues, and type of financing (Sichelman & Graham, 2010a). As might be expected, there is a stronger tendency by the VentureXpert companies – which are primarily funded by VC firms – to rate improving the chances of securing investment and liquidity events as more important compared with the Dun & Bradstreet companies – which are not generally venture-backed (Sichelman & Graham, 2010a). However, the overall order of the reasons listed is generally the same for each sample set as the aggregate presented in Fig. 2. However, when respondent D&B "population" firms are segmented by industry (Fig. 3), important differences arise, especially for companies competing in the software and Internet sector. For these firms, the importance of patenting for financing significantly lags, particularly as regards securing investment, and in fact patenting for "enhanced reputation" is cited as a significantly more important motive.

The next most important reasons for technology startups to file patents are associated with what are often called "defensive" and "strategic" motives, namely, preventing infringement lawsuits and improving negotiating positions, such as for cross-licensing (Fig. 2). Our finding that startup and early-stage firms rate these motives for patenting as "moderately important" is a novel finding, insofar as previous work had implicitly assumed – at least outside of the biotechnology industry – that small, young companies were not targeted in enforcement (litigation and licensing) activity at sufficiently high rates to justify using patents for defensive reasons (Graham & Sichelman, 2008). Although the early-stage technology firms in our sample indicate that these motives for patenting are less important than reported in the Carnegie-Mellon Survey (Cohen et al., 2000), our finding that startup companies may be engaging in sophisticated uses of patents for strategic and defensive purposes is nonetheless noteworthy and deserves further study.

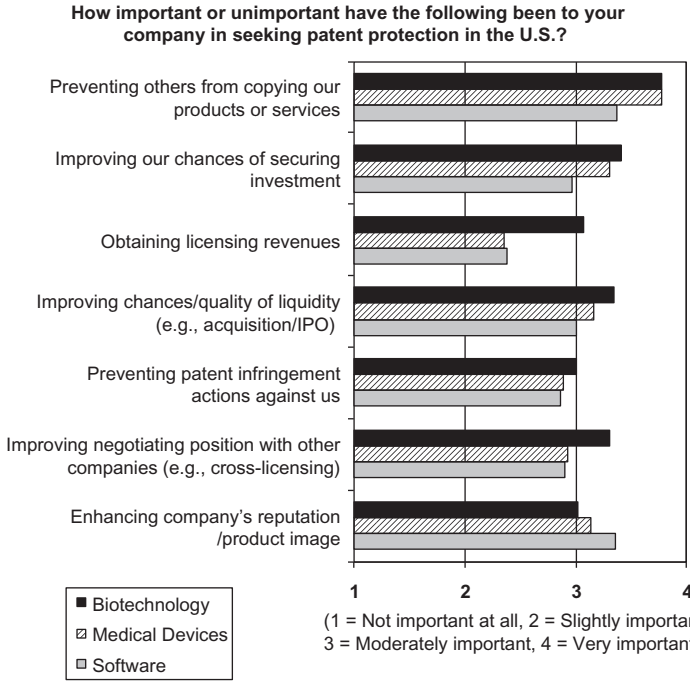



*Fig. 3.* Motivations for Seeking Patent Protection, by Industry. D&B Respondents only. Question Asked of Those Reporting Their Company had Filed for at Least One U.S. Patent (Averages Reported). *Source*: Reproduced from Graham et al. (2009).

Finally, among all startups, the importance of securing licensing revenue was rated significantly lower than other reasons, falling between "slightly important" and "moderately important" at the mean of all respondents. This finding might seem to conflict with the markets-for-technology view that small firms are more likely to license their patents because vertical specialization allows these firms to operate in upstream technology markets and provide technology inputs to firms operating in downstream product markets. Indeed, a survey of European patentees by Gambardella, Giuri, and Mariani (2005) showed that small firms are much more likely to patent to secure licensing revenue than larger ones. While our main findings show that licensing revenue is, in general, a comparatively unimportant consideration in startup patenting, some evidence from our study supports the view that the smallest of startup firms rely more on "patenting for

licensing" than do larger ones (Sichelman & Graham, 2008). And as we detail in the next section, firms in the biotechnology industry – which is often used as an exemplar of vertical specialization – are more likely to rate licensing income as an important reason to patent than are firms in other sectors we surveyed.

Consistent with the interviews we conducted, our results show significant inter-industry differences in the motives for filing patents. Examining responses from the population of companies (D&B) segmented by industry (Fig. 3), we find that the health and life science companies (biotechnology and medical devices) tend to cluster together in how they rate the importance of the different strategies. From a statistical standpoint, the averages presented in Fig. 3 for biotechnology and medical device respondents are indistinguishable, with the exception of "obtaining licensing revenues" and "improving negotiating position" which biotechnology firms rate as significantly more important motivations to file patents. Highlighting the industry distinctions, software and Internet firms' answers are all significantly different from the biotechnology and medical device firms, with the exception of "preventing patent infringement actions."

In particular, biotechnology and medical device firms rank preventing copying as nearly "very important" overall, while software firms place less emphasis on this motive (though still rating it between "moderately" and "very" important). The biotechnology and medical device companies also cite patenting to secure investment and to improve the chances and quality of a liquidity event as significantly more important motivations than do software firms. Biotechnology firms also place much greater emphasis on patenting to obtain licensing revenue than all other firms, including medical device firms, showing significant differences in their ratings overall. Interestingly, when software companies patent, executives report that "reputational" motives tend to be highly relevant, rated on average significantly more important than any other reason (excepting "preventing copying").

### *Why Entrepreneurs Choose against Patenting Major Innovations*

Having established the reasons for technology entrepreneurs selecting into using the patent system, we must explore the alternative. After all, a main contention of this chapter is that startups have a range of appropriability options, and patenting may not be the best choice in all circumstances. To investigate this issue more fully, we report here our BPS responses to a set

of questions concerning why these startups chose *not* to patent their major innovations.

A sizable percentage of respondents to the BPS held no patents or applications whatsoever (Graham et al., 2009). For the Dun & Bradstreet sample companies, nearly 60% had never filed for a patent – of software firms, roughly 75% had never filed, although an equal share of D&B biotechnology firms had filed. For the venture-backed, VentureXpert firms, only 18% had not filed – with 32% of software firms and 4% of biotechnology firms holding no patents or applications. Some of these non-filers – such as many of the small software service and "consulting" shops in our respondent sample – might simply not be innovating, at least in ways that are potentially patentable (although even non-patenting firms generally reported that innovation was important to their business strategies). On the other hand, firms that regularly patent may decide to forgo filing for some of their innovations. As discussed in Sichelman and Graham (2010b), because hindrances to startup patenting had not previously been thoroughly investigated with empirical data, we inquired of our sample firms why they decided not to patent their most recent major technology innovation.

In a survey of the literature, Graham and Sichelman (2008) investigated several reasons why startup firms may opt against patent protection, including (a) the belief that the technology is not patentable; (b) the high costs associated with prosecuting and enforcing the patent; (c) the perception, given the ability of competitors to reverse engineer, that patents may afford relatively weak protection; (d) the concern over technology disclosure; and (e) the availability of other forms of protection like copyrights and trademarks. Rather than simply asking our respondents who hold no patents to report on their motivations for choosing against patenting, we wanted to uncover the nuances underlying decisions to forgo patenting – even among those that were patenting their other inventions. As such, all respondents were asked whether the last major technology innovation they did not patent was a product or a process (or not), and what reasons motivated their company's decision to forgo a patent.

Fig. 4 shows the shares of reasons respondents reported for not patenting their company's last major technology innovation.[36] A major finding of our survey is that, when aggregating the responses of all technology startups, the cost of getting a patent is the most common reason cited for not patenting a major technology. In response to a follow-on question asking the respondent to list the single most important reason, lumping all respondents together yields the following ranking: cost of acquiring the patent

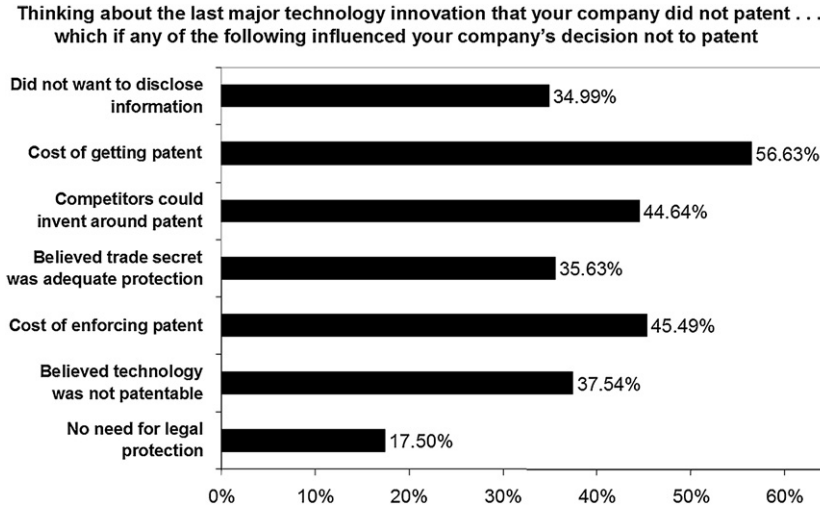



*Fig. 4.* Reasons for Startups to Forgo Patent Protection on Major Technologies. Respondents were Asked to Indicate All the Reasons that Applied (Share of Respondents Indicating that the Option Influenced the Decision is Reported). *Source*: Based on Graham et al. (2009).

(26.0%); did not believe technology was patentable (20.9%); did not want to disclose information (15.8%); ease of inventing around the patent (12.4%); cost of enforcing the patent (10.5%); no need for legal protection (7.3%); and believed that trade secret protection was adequate (7.0%). It is notable that, overall, our results are similar to those found in a Small Business Administration survey conducted in 1998 of small firms, which listed these same reasons at the top of small-business motivations for forgoing patenting (Cordes, Hertzfeld, & Vonortas, 1999).

Another major finding of our study reflects an industry distinction: the most important reason cited by biotechnology companies for not patenting is a reluctance to disclose information, while software companies most frequently cite to high patenting costs. In an effort to better understand the drivers of startups' choices to forgo patenting on their major innovations, we segmented the responses by industry and report the results in Fig. 5. We show that the most marked divergence occurs between biotechnology and software companies. Biotechnology firms are more than twice as likely to cite "disclosing information" as a reason to forgo patenting as are software firms (59% and 25%, respectively), and are more likely to believe that trade

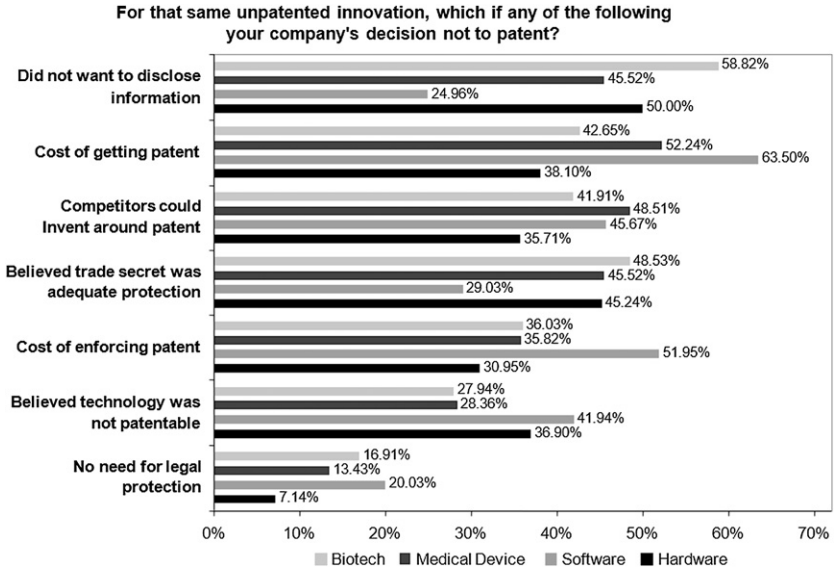



*Fig. 5.* Reasons Given for Not Seeking Patent Protection, by Industry. Respondents were Asked to Indicate All the Reasons that Applied (Share of Respondents Indicating that the Option Influenced the Decision is Reported). *Source*: Based on Sichelman and Graham (2010a).

secrets are an adequate means of protecting their innovations than are software firms (49% and 29%, respectively). A caveat is in order when interpreting these figures, however: the differences may be a consequence of the heightened likelihood of an unpatented biotechnology invention being a process technology, a possibility that we explore more thoroughly later in this chapter (see Fig. 7 below and accompanying text).

*Patenting Costs*

As is evident from Figs. 4 and 5, cost considerations in patenting loom large for startups, with the cost of prosecuting and the cost of enforcing the patent cited by more respondents than any other reason. By contrast, the difficulties and costs of acquiring and enforcing patents were less salient among larger firms surveyed in the Carnegie-Mellon study (Cohen et al., 2000), although even the relatively smaller firms answering that questionnaire tended to report a higher sensitivity to the costs of filing and enforcing patents.

Patenting costs can be quite high for startups. Answers to another BPS question revealed that the average out-of-pocket cost for a respondent firm to acquire its most recent U.S. patent, exclusive of R&D expenses, was over $38,000. This amount is significantly higher than costs reported by legal practitioners at the time the survey was taken, which varied from a low of $10,000 to a high of $20,000 (AIPLA, 2007). When asked to indicate the "most important" reason for not filing, more than one-third of the respondents selected either the cost of acquiring or enforcing the patent.

Notably, a non-trivial percentage of our respondents – about 10% – listed cost as the *only* barrier to filing for a patent from among the options we offered. This cost sensitivity may not be merely a result of more binding capital constraints, but may reflect a "startup cost premium." As stated previously, our respondents reported significantly higher patent prosecution costs than lawyers are reporting as the standard "market rate." In unstructured interviews we conducted, one executive at a venture-backed semiconductor firm stated that startups often pay significantly more than incumbents, because startups (i) tend to file for patents on inventions that are more important to the company's core business model than those filed by established firms, (ii) usually use outside, instead of in-house, counsel for patent prosecution, and (iii) often have difficulty monitoring outside counsel to limit overall costs.[37]

*Perceived Effectiveness of Patents*

Another set of reasons for forgoing patents relates to our respondents' perceptions about the effectiveness of the patent system. About 45% of respondents stated that the ability of competitors to design around a potential patent influenced their non-patenting decision, and nearly 38% of respondents reported not filing because they believed the innovation was not patentable (Fig. 4). These aggregated results reflect the high likelihood of software executives citing these reasons for non-patenting (Fig. 5), a somewhat counterintuitive result in view of criticisms that (i) the courts too often uphold patent claims of overly broad scope (Bessen & Meurer, 2009) and (ii) the U.S. Patent Office too frequently grants patents of questionable quality (Kesan & Gallo, 2006).

Based on the practice experience of one of the authors,[38] a non-trivial share of software innovators concerned with patent breadth may simply have been mistaken about how difficult it is to win from the USPTO patent claims of expansive scope during the 2000s (As we noted earlier, however, it is now more difficult to obtain software patents following the 2014 Alice v. CLS Bank decision). Supporting this contention is our finding that

the youngest firms, which are arguably less experienced with the Patent Office, are comparatively more likely to forgo filing because of the perceived ability of others to design around a patent (45% of software firms; Fig. 5). The same contention arguably holds given the large percentage (42%) of software firms that considered their innovations to be not patentable. It is possible, too, that among software engineers, a positive response to the "not patentable" option could refer to philosophic beliefs about what should be patentable and not objective beliefs about what was in fact patentable.[39]

Other considerations relating to these responses are worth commenting upon. First, startups citing more commonly to the ease of designing around a patent could simply reflect the distribution of "incremental" as opposed to "disruptive" innovation from these firms (Tidd, Bessant, & Pavitt, 2005). Specifically, incremental innovation is often harder to invent around, since doing so would require replicating other product components associated with the incremental innovation. Second, software patent applications, especially those in the "business method" area, have had relatively high rejection rates since at least 2006 (Lemley & Sampat, 2008). Third, Sichelman and Graham (2010a) found no significant differences in the perceived ability to invent around a patent or the patentability of the innovation-at-hand when segmenting respondent firms by the size of their patent portfolio, suggesting that companies with less experience in patenting share similar views with more patent-experienced firms concerning patenting thresholds and patent strength. In sum, just what share of startups are knowledgeable or ignorant about patentability standards and claim scope cannot be determined from our data.

*Disclosure Concerns*

The last set of reasons to forgo patenting revolves around the reluctance of firms to disclose their (trade) secrets. As shown in Fig. 4, respondents listed not wanting to disclose information as a reason for not patenting about 35% of the time. In this regard, if a firm makes reasonable efforts to prevent the disclosure of confidential information that provides it a commercial advantage, as we explained earlier, the firm is typically protected by trade secret law – so long as the company is aware the requirements for securing this protection.[40] We note that a nearly identical share of firms (36%) indicated that a reason not to file a patent was the adequacy of trade secret law. Moreover, that relatively few firms stated that they did not need any legal protection for their innovations indicates that trade secrecy law

may play an important role beyond the non-legal protection measures firms can take to ensure secrecy.

The relative frequency of these reasons tends to track with responses to the large-firm surveys, though preventing disclosure appears to be generally more of a reason to forgo patenting for large firms than for our respondents (Sichelman & Graham, 2010a). Part of this difference reflects our survey's heavy focus on software and Internet firms, for which reluctance to disclose was cited less commonly compared with other industries. This result may reflect the historically weak disclosure requirements at the U.S. Patent Office for software patents – particularly the ability of applicants to retain source code and other important know-how from patent applications (Maier, 1987). Or there may be other technology or labor-market characteristics at work that relate to information flows in the software and Internet space, including the notoriously short product life cycles in software, with market disclosure (putting a product on sale) happening prior to the usual 18-month patent application publication requirement.

*Non-Patenting Choices and Startup Funding Type*

Apart from industry, other company characteristics relate to startups' motivations in selecting against patenting their major innovations. For instance, when the respondent firms are segmented by data source (essentially, isolating a venture-funded sample), the venture-backed firms (VX) were more likely to cite disclosure concerns than were D&B "population" companies, the latter of which were generally smaller, younger, and not venture-funded (Fig. 6). At first blush, this finding seems puzzling since one might expect smaller firms to have greater concerns about larger competitors using their otherwise proprietary information. However, as the industry-segmented results show (Fig. 5), much of the reported difference stems from the biotechnology firms, which disproportionately report that process inventions are not patented (Sichelman & Graham, 2010a). Accordingly, the difference may reflect underlying dynamics in the types of technologies smaller, non-venture-backed firms choose to leave unpatented. Moreover, since D&B "population" firms consider legal protection less important overall than do their venture-backed VX counterparts (Graham et al., 2009), an explanation may be that disclosure is simply less of a competitive issue for the non-venture-backed sample.

Also notable in Fig. 6 is that D&B "population" firm respondents were significantly more likely to cite cost as the primary motivation to forgo patenting than their venture-backed VX counterparts. Indeed, as a group,

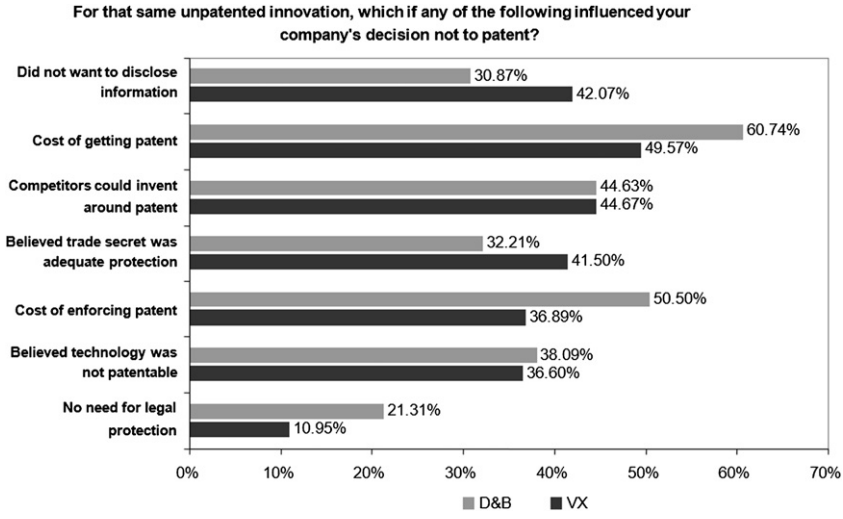



*Fig. 6.* Reasons Given For Not Seeking Patent Protection, by Sample Source. Respondents were Asked to Indicate All the Reasons that Applied (Share of Respondents Indicating that the Option Influenced the Decision is Reported). *Source*: Based on Sichelman and Graham (2010a).

while the VX firms listed filing costs (which include attorneys' fees) as the main reason for not patenting, the costs of enforcement ranked lower (below "designing around the patent"). Such differences in sensitivity to cost may be attributable to the greater capitalization of venture-backed firms. Analysis completed in Sichelman and Graham (2010a) shows that when firms are segmented by annual revenue, statistically significant differences appear between the highest and lowest quartile in terms of the barrier to patenting presented by prosecution costs, and above and below the median in terms of patent enforcement costs (in each case, higher revenue firms showed less sensitivity). Additional analysis in Sichelman and Graham (2010a) dividing the sample by age of firm since founding (0–4 years vs. 8–10 years) shows that the older firms were significantly less sensitive to the cost of acquiring and enforcing patents than were the younger companies. In addition, comparing Figs. 1 and 5 demonstrates that firms operating in technologies where patents are considered relatively effective at appropriating value from innovation (i.e., biotechnology and medical devices) cited filing and enforcement costs as disincentives to patenting

significantly less often than software and Internet companies (where patents are considered less effective protection).

*Non-Patenting Choices and Innovation Types*

Whether the innovation is a product or process raises other interesting differences regarding the disclosure of proprietary information. We segmented our data by whether the respondent indicated that the last invention for which they chose to forgo patent protection was a product or a process (Fig. 7). As expected, trade secrecy was cited significantly more commonly as adequate protection for process inventions as compared to product inventions (although the latter still had a substantial number of respondents citing to the adequacy of trade secret). This result likely stems from the characteristic that processes, especially internal ones, are generally easier to keep secret than products (Teece, 1986). For similar reasons, respondents were significantly more likely to cite a concern over disclosure for processes as against products in patent applications. It is also notable in Fig. 7 that concerns over costs appear to be more germane for startups that chose to forgo patent protection on (their last) product innovation, although we suspect this result is driven by a selection effect: The

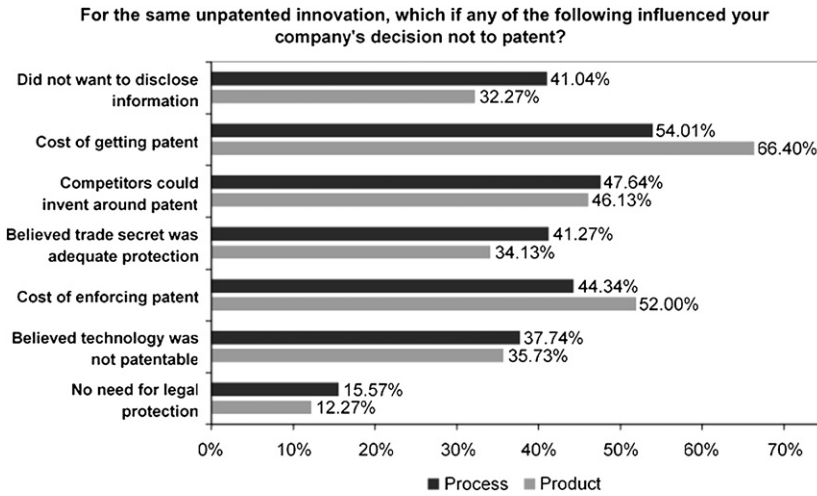



*Fig. 7.* Reasons Given For Not Seeking Patent Protection, by Firm Innovation Focus. Respondents were Asked to Indicate All the Reasons that Applied (Share of Respondents Indicating that the Option Influenced the Decision is Reported). *Source*: Based on Sichelman and Graham (2010a).

types of firms we have in our sample (skewed toward software), as well as those choosing not to patent (for which patenting may be less useful for a host of reasons), will tend to have their answers to this question more commonly represented. We therefore may have answers that are skewed toward making cost considerations look particularly important for product innovators. However, since our previous results show that cost is an important deterrent to startup patenting, these answers can be read to give at least qualitative support for other information reported in the BPS.

The results in Fig. 7 also demonstrate that the cost of filing is a particularly salient factor for those companies that chose to forgo patenting on a product technology, with nearly one-third (32%) citing this reason as the most important factor. When firms choose to leave process technologies unpatented, they are most likely to cite three reasons about equally: reluctance to disclose information (24%), a belief that the technology was unpatentable (22%), and, again, the cost of filing (21%). As such, because biotechnology companies were more likely to report that their last unpatented innovation was a process innovation, part of the inter-industry differences described earlier may be explained by underlying differences in the types of technologies respondents were contemplating patenting.

## CONCLUSION

The sum of recent scholarship suggests that scholars, teachers, and students interested in the commercialization of technology should take a much more nuanced view of the mechanisms available to entrepreneurs for capturing value from their innovations. Managers in companies, and those working with scientists and engineers in the labs, report that patents are not in all circumstances most effective when compared with the other tools available to the company for capturing and sustaining supernormal profits. Prior surveys (Cohen et al., 2000; Levin et al., 1987) had suggested "other legal" means could be effective, and the BPS, by specifically asking technology startup executives about copyright, trademark, and secrecy, validated this view. Other scholarly research has demonstrated that non-patent protections continue to be useful methods for competing effectively in the knowledge economy (Graham & Mowery, 2003, 2004; Graham & Somaya, 2006).

This chapter has argued that scholars and teachers of technology entrepreneurship need to take a broader view of IP protections, beyond mere patenting. When considering IP in the context of technology

commercialization, those in the university often think first of the Bayh-Dole Act, and of patents. In the foregoing text, we introduced in a technology commercialization setting the relevance of non-patent forms of IP, such as copyright, trademark, and trade secret, and offered a framework for thinking about these protections, and the opportunities they offer to entrepreneurs both inside and outside the university.

While patents are normally the "default" position when we think about protecting technologies and profiting from them, we show that since patents have been shown in large-company surveys to be among the *least* important means of capturing value from innovation (Cohen et al., 2000; Levin et al., 1987), we would do better by expanding the way in which we think about protecting and capturing value from innovations. The chapter highlights how the importance of non-patent forms of IP to entrepreneurs is revealed by answers startup managers provided in the *BPS* (Graham et al., 2009).

We also argue in this chapter against another preconception (held by many scholars, teachers, and students) that IP protections act as substitutes for one another. We suggest the field should take a more nuanced view, abstracting away from our focus on single "inventions" to a broader view of innovation as a dynamic and complex process – one that allows innovators to employ different types of IP together, in ways that may indeed be complementary (Graham & Sichelman, 2008; Graham & Somaya, 2006; Simon & Sichelman, 2016). We also offer evidence from the *BPS* that entrepreneurs are examining the problem in exactly this sense, and are using patents, copyrights, trade secrets, and trademarks together, throughout the innovation process, in ways that appear to complement one another. In this regard, we suggest that scholars and teachers of commercialization ought to carefully internalize these views. By doing so, we would do a more thorough job of opening the world to our audiences by helping them recognize the complementarities that exist between different forms of IP and the importance of non-patent forms of IP in the technology commercialization process.

## NOTES

1. 94 Stat. 3015 (1980).
2. Collectors of valuable information can prevent competitors from using the information. *International News Service v. Associated Press*, 248 U.S. 215 (1911).

3. Protection is available for software embodied in semiconductor chips – so-called mask works. *E.F. Johnson v. Uniden Corp. of America*, 653 F. Supp. 1485 (D. Minn. 1985).

4. Copyright protection, like patent protection, emanates from the U.S. Constitution, in Article I, Section 8, but the protection and its parameters must be legislated by Congress.

5. 17 U.S.C.A. § 102 (2014).

6. *Ibid.*

7. 17 U.S.C.A. § 106, § 302 (2014).

8. 37 C.F.R. Part 201.3 (2014).

9. 17 U.S.C.A. §410 (2014).

10. 17 U.S.C.A. §411 (2014).

11. The holder of a registered copyright is entitled to the recovery of attorney fees and statutorily defined damages, including those for willful infringement, only for the period after registration. Ordinarily, the owner cannot collect these damages for the period between the time of publication and registration of the copyright, but the law offers an incentive for registering *early*: damages are available from the date of publication *only* if the owner registers the copyright within 3 months of publication of the work. 17 U.S.C.A. §§412, 504, 505 (2014).

12. 17 U.S.C. §§ 101, 117 (as amended 1980), included commentary. For a more complete discussion, see Menell (1987).

13. See *Computer, Inc. v. Franklin Computer Corp.*, 714 F.2d 1240 (3rd Cir. 1983).

14. See, for example, the series of Lotus against Borland spreadsheet copying cases. *Lotus Development Corp. v. Borland Int'l, Inc.*, 788 F. Supp. 78 (D. Mass. 1992) (finding the *Quattro* spreadsheet a virtual copy of Lotus' menu structure); *Lotus Development Corp. v. Borland Int'l, Inc.*, 799 F. Supp. 203 (D. Mass. 1992); *Lotus Development Corp. v. Borland Int'l, Inc.*, 831 F. Supp. 202 (D. Mass. 1993); *Lotus Development Corp. v. Borland Int'l, Inc.*, 831 F. Supp. 223 (D. Mass. 1993); *Lotus Development Corp. v. Borland Int'l, Inc.*, 49 F.3d 807 (1st Cir. 1995), *aff'd* 516 U.S. 233 (1996).

15. 134 S. Ct. 2347 (2014); See Jones (2014).

16. *Reddy Comm. v. Environmental Action Fnd.*, 477 F. Supp. 936 (D.D.C. 1979).

17. 15 U.S.C.A. §1051 *et seq.* (2014).

18. 15 U.S.C.A. §§1053, 1054, 1127 (2014). For instance, United Parcel Service in 2015 was the registered owner of the distinctive shield "UPS" trademark (U.S. Reg. No. 3160056), the "We Love Logistics" service mark (U.S. Reg. No. 4553000), and the distinctive "chocolate brown" color mark for vehicles and uniforms used in delivery (U.S. Reg. Nos. 2901090, 2159865, and 2131693).

19. 15 U.S.C.A. §1051 (2006).

20. 15 U.S.C.A. §§1072 (2006).

21. *Zatarian's, Inc. v. Oak Grove Smokehouse, Inc.*, 698 F.2d 786 (5th Cir., 1983).

22. Scarves by *Vera v. Todo Importa Ltd., Inc.*, 544 F.2d 1167 (2nd Cir. 1976).

23. However, if the holder fails to "use" the mark in commerce or fails to "police" the mark by taking action against others' diluting uses, the mark may be lost. It may even enter the public domain through "genericide." Famous examples of the latter include aspirin and cellophane.

24. *Metallurgical Industries, Inc. v. Fourtek, Inc.*, 790 F.2d 1195 (5th Cir., 1986); UTSA (1985).

25. *E.I. DuPont de Nemours & Co. v. Christopher*, 431 F.2d 1012 (5th Cir., 1970).

26. 416 U.S. 470.

27. *United States v. Pilkington, PLC*, 1994-2 Trade Cas. (CCH); 70,842, 1994 WL 750645 (D. Ariz. 1994).

28. The Department of Justice considered this use of trade secret as the maintenance of unwarranted monopoly power. *See ibid.*

29. Much of the controversy surrounded philosophical (normative) questions concerning whether universities should shift their focus away from "blue sky" to more applied research, and evaluations (positive) of whether universities and their stakeholders were better or worse off for having made the investments in patent-based technology transfer facilitated by the Bayh-Dole regime. The evidence on the latter issue is mixed (Mowery & Sampat, 2001, 2004).

30. Patent protection is considered a relatively strong type of protection when compared to other types of IP since, for instance, patent protection will block third-party uses even if the technology was invented independently, unlike trade secret protection. However, it is worth noting that copyright and trade secret protections can have a much longer life than patent protection and are considerably less expensive to secure than patents.

31. D&B is a leading business credit-reporting and information source in the United States holding over 140 million business records worldwide in 2008. The company reported that "87 percent of D&B's U.S. active file contains businesses with 10 or fewer employees." Dun & Bradstreet, Facts & Figures, http://www.dnb.com/us/about/db_database/dnbstatistics.html (last visited May 24, 2009).

32. *VentureXpert* catalogs venture-backed companies and provides information about executives, funding, and deal attributes (Kaplan, Strömberg, & Sensoy, 2002).

33. Our list included all of D&B's biotechnology companies (642 in total) assigned to NAICS 541711, all medical-device companies (1,048) assigned to SIC 3841, and a random sample of its listed software companies (8,810) assigned to SIC codes 7371, 7372, 7373, and 7379. For the *VentureExpert* database, we relied on the technology classification provided by Thomson for each company.

34. When we use the term "significant" in this chapter, we are referring to statistically meaningful differences to confidence levels of at least 95%.

35. We define "product innovators" here as those that rated product innovation as "very important" but all other types of innovation as less important. Similarly, we define "process innovators" as those that rated process innovation as "very important" but all other types of innovation as less important.

36. This question was answered by respondents regardless of whether they had filed for a patent. Because respondents could select several reasons, the percentages for each do not sum to 100%.

37. Telephone interview with an unidentified semiconductor company startup executive (February 20, 2009).

38. Co-author Sichelman practiced patent law, litigating patent disputes in the software industry for a number of years before becoming a professor and researcher.

39. The former explanation finds support in results from a prior attitudinal survey (Samuelson & Glushko, 1990).
40. See, for instance, the UTSA §§ 1–2 (1985).

## REFERENCES

American Intellectual Property Law Association [AIPLA]. (2007). *Results of the economic survey*. Washington, DC: AIPLA.

Arora, A. (1995). Licensing tacit knowledge: Intellectual property rights and the market for know-how. *Economics of Innovation and New Technology*, *4*, 41–59.

Arora, A. A., Fosfuri, A., & Gambardella (2004). *Markets for technology: The economics of innovation and corporate strategy*. Cambridge, MA: MIT Press.

Arrow, K. (1962). Economic welfare and the allocation of resources for invention. In *The rate and direction of inventive activity: Economic and social factors* (pp. 609–626). Princeton, NJ: Princeton University Press.

Bessen, J., & Meurer, M. J. (2009). *Patent failure: How judges, bureaucrats, and lawyers put innovators at risk*. Princeton, NJ: Princeton University Press.

Boldrin, M., & Levine, D. K. (2013). The case against patents. *Journal of Economic Perspectives*, *27*(1), 3–22.

Ceccagnoli, M., & Rothaermel, F. (2016). Appropriability strategies to capture value from innovation. In M. C. Thursby (Ed.), *Technological innovation generating economic results – the evolving landscape* (Vol. *26*). Advances in the Study of Entrepreneurship, Innovation and Economic Growth. Bingley, UK: Emerald Group Publishing Limited.

Cockburn, I. M., & MacGarvie, M. (2009). Patents, thickets and the financing of early-stage firms: Evidence from the software industry. *Journal of Economics and Management Strategy*, *18*(3), 729–773.

Cohen, W. M., Nelson, R. R., & Walsh, J. (2000). *Protecting their intellectual assets: Appropriability conditions and why U.S. manufacturing firms patent (or Not)*. NBER Working Paper 7552.

Cordes, J. J., Hertzfeld, H. R., & Vonortas, N. S. (1999). *A survey of high technology firms*. Washington, DC: Small Business Administration. Retrieved from http://sba.gov/advo/research/rs189tot.pdf

Friedman, D. D., Landes, W. M., & Posner, R. A. (1991). Some economics of trade secret law. *Journal of Economic Perspectives*, *5*(1), 61–72.

Gambardella, A., Giuri, P., & Mariani, M. (2005). *The value of European patents: Evidence from a survey of European inventors* – Report. Retrieved from http://tinyurl.com/zdw7x96

Graham, S. J. H., Merges, R. P., Samuelson, P., & Sichelman, T. (2009). High technology entrepreneurs and the patent system: Results of the 2008 Berkeley patent survey. *Berkeley Technology Law Journal*, *24*(4), 255–327.

Graham, S. J. H., & Mowery, D. C. (2003). Intellectual property protection in the U.S. software industry. In W. M. Cohen & S. Merrill (Eds.), *Patents in the knowledge-based economy*. Washington, DC: National Academies Press.

Graham, S. J. H., & Mowery, D. C. (2004). Submarines in software? Continuations in U.S. software patenting in the 1980s and 1990s. *Economics of Innovation and New Technology*, *14*, 443–456.

Graham, S. J. H., & Sichelman, T. (2008). Why do start-ups patent? *Berkeley Technology Law Journal*, *23*(3), 1063–1097.

Graham, S. J. H., & Sichelman, T. (2011). Patenting by high tech entrepreneurs, in creativity. In S. Ghosh & R. Malloy (Eds.), *Law and entrepreneurship*. Cheltenham: Edward Elgar.

Graham, S. J. H., & Somaya, D. (2006). *Vermeers and rembrandts in the same attic: Complementarity between copyright and trademark in software firms' IP leveraging strategies*. SSRN Working Paper 887484.

Griliches, Z. (1990). Patent statistics as economic indicators: A survey. *Journal of Economic Literature*, *28*, 1661–1707.

Hall, B. H. (2005). Exploring the patent explosion. *Journal of Technology Transfer*, *30*(2), 35–48.

Hall, B. H., Jaffe, A. B., & Trajtenberg, M. (2001). *The NBER patent citation data file: Lessons, insights and methodological tools*. NBER Working Paper 8498.

Holbrook, T. (2016). The America invents act: A new patent system for the 21st century. In M. C. Thursby (Ed.), *Technological innovation generating economic results – the evolving landscape* (Vol. *26*). Advances in the Study of Entrepreneurship, Innovation and Economic Growth. Bingley, UK: Emerald Group Publishing Limited.

Horstman, I., MacDonald, G., & Slivinsky, A. (1985). Patents as information transfer mechanism. *Journal of Political Economy*, *93*, 837–858.

Hsu, D., & Ziedonis, R. (2007). *Patents as quality signals for entrepreneurial ventures*. Paper presented at the Druid Summer Conference 2007, Copenhagen Business School, Copenhagen 18–20 June 2007.

Jones, A. (2014). Courts nix more software patents: Decisions follow supreme court ruling on intellectual-property protection. *Wall Street Journal, Law Journal*, September 21.

Kaplan, S. N., Strömberg, P., & Sensoy, B. A. (2002). *How well do venture capital databases reflect actual investments?* SSRN Working Paper 939073.

Kesan, J. P., & Gallo, A. A. (2006). Why "Bad" patents survive in the market and how should we change?—The private and social costs of patents. *Emory Law Journal*, *55*, 61–97.

Lemley, M. A., & Sampat, B. (2008). Is the patent office a rubber stamp? *Emory Law Journal*, *58*, 181–227.

Levin, R. C., Klevorick, A. K., Nelson, R. R., & Winter, S. G. (1987). Appropriating the returns from industrial research and development. *Brookings Papers on Economic Activity*, *3*, 783–820.

Maier, G. J. (1987). Software protection—Integrating patent, copyright and trade secret law. *Journal of the Patent & Trademark Office Society*, *69*, 151–192.

Mann, R. J. (2005). Do patents facilitate financing in the software industry? *Texas Law Review*, *83*(4), 961–1030.

Menell, P. (1987). Tailoring legal protection for computer software. *Stanford Law Review*, *39*, 1329–1372.

Meurer, M. J. (2008). Inventors, entrepreneurs, and intellectual property law. *Houston Law Review*, *45*, 1201–1281.

Moser, P. (2005). How do patent laws influence innovation? Evidence from nineteenth-century world's fairs. *American Economic Review*, *95*(4), 1214–1236.

Mowery, D. C., & Sampat, B. N. (2001). University patents and patent policy debates in the USA, 1925–1980. *Industrial and Corporate Change*, *10*(3), 781–814.

Mowery, D. C., & Sampat, B. N. (2004). The Bayh-Dole act of 1980 and university–industry technology transfer: A model for other OECD governments? *Journal of Technology Transfer*, *30*(1–2), 115–127.

National Commission on New Technological Uses of Copyrighted Works [CONTU]. (1979). Final Report. USGPO: Washington, DC.

Plant, A. (1934a). The economic theory concerning patents for inventions. *Economica*, *1*(1), 30–51.

Plant, A. (1934b). The economic aspects of copyright in books. *Economica*, *1*(2), 167–195.

Rector, A. M., Sandefur, B., Ceccagnoli, M., Clendenin, M., & Hallenborg, L. (2016). Introduction to legal means for protecting intellectual property. In M. C. Thursby (Ed.), *Technological innovation generating economic results – the evolving landscape* (Vol. 26). Advances in the Study of Entrepreneurship, Innovation and Economic Growth. Bingley, UK: Emerald Group Publishing Limited.

Rumelt, R. P. (1984). Towards a strategic theory of the firm. In R. B. Lamb (Ed.), *Competitive strategic management*. Englewood Cliffs, NJ: Prentice-Hall.

Samuelson, P. (1984). CONTU revisited: The case against copyright protection for computer programs in machine-readable form. *Duke Law Journal*, *1984*, 663–752.

Samuelson, P., & Glushko, R. J. (1990). Survey on the look and feel lawsuits. *Communications of the ACM*, *33*(5), 483–495.

Schumpeter, J. A. (1934). *The theory of economic development; an inquiry into profits, capital, credit, interest, and the business cycle*. Cambridge, MA: Harvard University Press.

Sichelman, T. (2010). Commercializing patents. *Stanford Law Review*, *62*(2), 341–413.

Sichelman, T., & Graham, S. J. H. (2010a). Patenting by entrepreneurs: An empirical study. *Michigan Telecommunications and Technology Law Review*, *17*, 111–180.

Sichelman, T., & Graham, S. J. H. (2010b). Why do entrepreneurs patent? In R. Litan (Ed.), *Handbook of law, innovation and economic growth*. Cheltenham: Edward Elgar.

Simon, B., & Sichelman, T. (2016). *Data-generating patents*. Unpublished working paper.

Teece, D. J. (1986). Profiting from technological innovation: Implications for integration, collaboration, licensing and public policy. *Research Policy*, *15*(6), 285–305.

Thoma, G., Torrisi, S., Gambardella, A., Guellec, D., Hall, B. H., & Harhoff, D. (2010). *Harmonizing and combining large datasets–An application to firm-level patent and accounting data*. NBER Working Paper 15851.

Tidd, J., Bessant, J., & Pavitt, K. (2005). *Managing innovation: Integrating technological, market and organizational change* (3rd ed.). Chichester: Wiley.

University of Washington. (2007). University of Washington handbook. In *Patent, invention, and copyright policy* (Chapter 7). Seattle, WA: University of Washington.

World Intellectual Property Organization [WIPO]. (2013). *World intellectual property report 2013: Brands–reputation and image in the global marketplace*. Geneva: WIPO.